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1 Network Architecture

The backbone for this network architecture is the *MobileNetV3-small* pretrained model from the *torchvision.models* library. This backbone is split up into three sequential parts: the low-level features (fine-grained details), the mid-level features (intermediate representations), and the high-level features (semantic information).

A context module is added after the backbone with three branches each with dilation rates of 1, 6, and 12 respectively. It is projected to 128 channels and includes a 2D dropout with $p=0.1$.

Depthwise separable convolution is used for the low and mid level features. This consists of a depthwise convolution, then a pointwise convolution, followed by batch normalization and a ReLU activation.

After the depthwise separable convolutions, each of the three feature levels are upsampled to the same size as the low-level features using bilinear interpolation, and then concatenated together. They are then sent through a decoder convolution block which uses two 3×3 convolutions, batch normalization and a ReLU function.

Finally, the image is upsampled to the original size using bilinear interpolation, at which point dropout with $p=0.1$ is applied and a 1×1 convolution which maps the output to one of 21 classes.

2 Knowledge Distillation